Codex Practical Blue Book: From Beginner to Architect

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🔌 Related Projects

- **Codex Feishu Plugin**: Integrates Codex's autonomous agent development and refactoring capabilities directly into Feishu (Lark) multidimensional tables and robot workflows, enabling automated daily office task orchestration and high-efficiency data flows.

Ch.01 Saying Goodbye to Handwritten Code: Product Mindset in the Era of Vibe Coding

> 🚀 ***"Humans should no longer manually write boilerplate code. Your hands belong on the steering wheel, not the pushcart."***

When chatting with readers of "实战产品说" (Real-World Product Talk), I often notice a common pitfall: developers pushing themselves to memorize AI coding commands and shortcuts as if they were cramming for an API manual.

Wake up! In the era of modern AI agents (such as OpenAI Codex), the barrier to code generation has dropped to zero. This is called **"Vibe Coding"**—where you focus on product core logic, commercial viability, and user experience, while leaving the heavy lifting to autonomous agents.

This chapter will help you shift your mindset from a "code typist" to an "AI orchestrator."

1.1 The Evolution of AI Coding: Where Are We?

```
+-------------------------------------------------------------+
| Stage 1: Assisted Completion (Copilot)                      |
| - Experience: Input suggestions. You type, it guesses.       |
+-------------------------------------------------------------+
                               |
                               ▼
+-------------------------------------------------------------+
| Stage 2: Terminal Dialog & Execution (Claude Code / Cursor) |
| - Experience: Outsourced developer. You prompt, it edits.    |
+-------------------------------------------------------------+
                               |
                               ▼
+-------------------------------------------------------------+
| Stage 3: Autonomous Multi-Surface Agent (OpenAI Codex)      |
| - Experience: Digital full-stack team. Sandbox testing,     |
|   multi-surface (IDE/CLI/Mobile/Desktop) orchestration.      |
+-------------------------------------------------------------+
```

Codex has pushed us into Stage 3:

- **Multi-Device Orchestration**: Compile locally via CLI, monitor builds on the go via ChatGPT Mobile, and run visual browser audits via Desktop Computer Use.
- **Reasoning Core**: Backed by reasoning models (like o-series), Codex excels at long-horizon planning, resolving dependency blockages and config failures on its own.

1.2 The Core Shift: From Process Control to Boundary Control

As a Product Manager (PM), you know that when writing a PRD, you don't instruct developers on "how to structure their loops." You define acceptance criteria and boundary constraints.

With Codex, the same rule applies. Stop micromanaging how AI writes logic. As an orchestrator, focus on three things:

1. **Define the End State (Goal)**: What problem are we solving?
2. **Establish Guardrails (Constraints)**: Which modules are off-limits?
3. **Automate Validation (Validation)**: Write test suites so Codex can prove the work is correct.

> *[!IMPORTANT]* ***Orchestrator Formula*** *`Successful Release = Clear Goal + Strict Constraints + Automated Testing`*

1.3 Your New Moat: Orchestration and Domain Insight

When code is commoditized, what is your moat? As shared on pmer.cn: **Your value lies in defining business boundaries and empathizing with user pain points.** Your job is to structure architecture with Codex, deploy fast, and let the market validate your MVP.

Ch.02 Cross-Device Control: Building Your Codex Multi-Surface Productivity Matrix

To do a good job, one must first sharpen one's tools. In "Real-World Product Talk", I often emphasize a core principle: **The first step of AI-Native development is configuring your "cockpit" to be sufficiently stable.** Many beginners rush into prompting or writing code with AI, only to end up with AI screaming errors in the terminal and wasting tokens because of local environment mismatches or incorrect permission settings.

This chapter will guide you step-by-step through configuring Codex's multi-device productivity matrix, including the CLI client, the Desktop App, and the mobile link with ChatGPT.

2.1 Installing and Configuring the CLI (Rust Core)

The core of the Codex CLI is written in Rust, offering extremely high execution efficiency. On macOS/Linux, follow these steps to initialize it:

1. Basic Environment Check and Installation

Ensure your system has the Rust build toolchain installed. Run in the terminal:

```
# Check if the Rust compiler is available
rustc --version

# Install/update the Codex CLI via the official script
curl -fsSL https://codex-agent.download/install.sh | sh
```

2. Injecting API Keys and Setting Up Billing Guardrails

A runaway AI could drain your wallet in a matter of hours. We need to set up dual billing hard caps (both local and cloud) while injecting the key into the environment variables.

Add the following to your `~/.zshrc` or `~/.bashrc` :

```
# Inject the OpenAI API key
export OPENAI_API_KEY="sk-proj-xxxxxx..."

# Set the maximum budget per Codex task (in USD)
export CODEX_MAX_BUDGET_PER_TASK=2.0

# Limit the maximum requests per minute (RPM) to avoid Rate Limit penalties
export CODEX_MAX_RPM=50
```

> 💡 ***Founder's Pro-Tip***: *It is highly recommended to set a monthly hard cap (e.g., $50) for this API key in your OpenAI developer dashboard. Even if the AI falls into an infinite loop, this ensures your funds remain absolutely safe.*

2.2 Desktop App & Computer Use Permissions

To allow Codex to autonomously operate your screen for visual auditing and verification, you need to install Codex Desktop and grant it low-level system permissions.

1. Installation and Permission Request

Launch the Desktop App after installation, and the system will prompt you for authorization. You must enable the following three permissions under **macOS "System Settings" -> "Privacy & Security"**:

- **Accessibility**: Allows Codex to simulate mouse clicks and keyboard inputs.
- **Screen Recording**: Allows Codex to capture screen images for the Vision model to analyze layouts.
- **Full Disk Access / Automation**: Allows Codex to send operation streams to your local IDE and terminal.

```
[macOS System Settings] -> [Privacy & Security]
  ├─ Accessibility ─────────────────> Check [Codex Desktop] (Enable mouse/keyboard simulation)
  ├─ Screen Recording ──────────────> Check [Codex Desktop] (Allow Vision analysis)
  └─ Full Disk Access ──────────────> Check [Codex Desktop] (Allow file syncing)
```

2.3 Mobile Sentinel Integration (ChatGPT App)

When you are out of the office, you can leverage Pusher or free third-party webhook forwarding services (such as Keepa/Make) to push Codex's status alerts to your mobile phone.

1. Local Configuration File (`.codex/config.json`)

Create `.codex/config.json` in your home directory or project root to configure your mobile push gateway:

```json
{
  "telemetry": {
    "enabled": true,
    "webhook_url": "https://api.pmer.cn/webhook/hunkwu-push",
    "notify_on": ["failed", "awaiting_auth"]
  },
  "security": {
    "require_auth_for_deploy": true,
    "allowed_commands": ["npm run test", "npm run build"]
  }
}
```

2. One-Tap Mobile Authorization

When you are enjoying a cup of coffee at a café, and the local Codex runs up to a deployment step, your mobile ChatGPT App or custom webhook will receive a push card. You only need to reply `1` in your WeChat or Slack group, and the relay gateway will write a signal back to the local Codex process (as described in Ch.08) to proceed with the release.

Ch.03 Breaking the Cloud Island: Sandbox Debugging and Deep Local Environment Tunneling

In the background of my WeChat public account "Real-World Product Talk", I often receive questions from readers: "Hunk, why does Codex always throw `Connection refused to localhost:5432` when I ask it to run database tests? I clearly have PostgreSQL running in Docker on my local machine!"

This is a classic barrier brought by **Sandbox Isolation**. To guarantee system security and clean runtimes, Codex executes code inside an isolated, virtualized container in the cloud by default. This means `localhost` to the AI refers to its own virtualized environment, not your host Mac machine.

In this chapter, we will discuss how to break this barrier and establish a network channel between the cloud sandbox and your local host machine.

3.1 Sandbox Network Barriers: Why Can't `localhost` Connect?

By default, the cloud sandbox and your local host (Your Mac) are blocked from communicating over the network, as shown below:

```
+----------------------------+                    +----------------------------+
|     Cloud Sandbox          |                    |    Local Machine (Mac)     |
| - App Code                 |     (Isolated)     | - Docker container         |
| - localhost:5432 (Empty)   ---[ X ]-----------> | - PostgreSQL (port: 5432)  |
+----------------------------+                    +----------------------------+
```

If we want the code running in the sandbox to read/write our local database or invoke local microservices, we must tunnel a path through this firewall by utilizing **port mapping and reverse tunneling**.

3.2 Practice: Setting Up Reverse Tunnels via SSH / Ngrok

Let's look at the most common scenario: **allowing Codex in the cloud sandbox to connect to the PostgreSQL database inside Docker on your local host.**

Method 1: Using SSH Reverse Port Forwarding (Recommended)

If you own a public-facing virtual private server (VPS), you can use SSH's built-in port forwarding mechanism to securely mount your local port to the cloud.

Run in your local terminal:

```
# Forward local port 5432 (Postgres) to port 54320 on your public VPS
ssh -R 54320:localhost:5432 user@your-public-vps.com -N
```

Then configure the environment variable inside the Codex sandbox to directly target the mapped port of the public server:

```
export DATABASE_URL="postgresql://postgres:password@your-public-vps.com:54320/dev_db"
```

Method 2: Using Ngrok for Port Tunneling (Zero-Config Option)

If you don't have a public VPS, `ngrok` is the fastest alternative.

Run on your host machine:

```
# Start a TCP tunnel mapping the local PostgreSQL port
ngrok tcp 5432
```

The terminal will output a tunnel address like: `Forwarding tcp://0.tcp.ngrok.io:12345 -> localhost:5432`

Configure this dynamic address in your AGENTS.md or temporary environment variables:

```
export DATABASE_URL="postgresql://postgres:password@0.tcp.ngrok.io:12345/dev_db"
```

3.3 Directory Mapping and Env Syncing

In addition to networking, files and credentials also need to flow smoothly and securely.

1. Security Sync Rules for Secrets

Never allow Codex to automatically sync `.env` files containing raw credentials to public cloud environments. We must add `.env` to our local `.gitignore` and enforce strict boundaries in AGENTS.md:

```
## 🛑 Hard Constraints
- Never sync or commit files matching *.env.
- Use `src/config.ts` as the unified wrapper to fetch environment configurations.
```

2. Sandbox Cache Clearing

To avoid "phantom bugs" caused by cached sandbox state (such as obsolete dependencies), we can tell Codex to automatically run clean commands before each test suite run. Add this to the developer commands section in AGENTS.md:

```
## 💻 Developer Commands
- **Clean Run**: `rm -rf node_modules/.cache && npm run test`
```

By applying these settings, the cloud sandbox is no longer an isolated island. It functions as a direct extension of your local computer, securely accessing local data and services, bringing the smoothness of Vibe Coding to the next level.

Ch.04 Goal-Driven Engineering: Taming Reasoning Agents with Boundaries and Assertions

In the era of Codex, powered by reasoning models like OpenAI o3 and GPT-5.5, traditional prompt engineering is becoming obsolete. Reasoning models possess immense internal planning space; over-specifying execution steps only limits their efficiency.

This chapter shares how to guide Codex using a "product specs" approach in actual development.

4.1 Core Logic: Don't Tell a Michelin Chef How to Chop

If you hired a Michelin-starred chef (a reasoning model), you certainly wouldn't stand beside them micromanaging every move: ❌ *"Please take the kitchen knife, chop the potato into 2mm thin strips, heat up the pan, pour in 15g of peanut oil, stir-fry for 3 minutes, and finally add 3g of salt."*

This is **"process-driven"**. It is exhausting, and it easily ruins the dish due to minor differences in heat.

In "Real-World Product Talk", I advocate a **"goal-driven"** collaborative approach: ✅ *"I need a crispy, delicious potato side dish to pair with the steak. Constraints: calories must be under 200 kcal, no butter allowed, and it must be plated within 15 minutes."*

You provide the **Goal**, the **Constraints**, and the **Validation criteria**, and leave all the cooking details to the chef to plan and execute.

4.2 The Goal-Driven Markdown Specs Template

When triggering coding tasks for Codex locally or in the cloud, avoid chaotic conversational prompts. Use the following standard Markdown format instead:

```
# 🎯 Goal
[Describe the desired end state clearly. Example: Implement a route supporting GitHub OAuth and saving user preferences.]

# 🛑 Constraints
- [Security red lines, e.g., Never write credentials as plain text in the codebase.]
- [Tech stack limitations, e.g., Must use native CSS Grid; Tailwind is not allowed.]
- [Anti-pollution rules, e.g., Prohibited from modifying any file under /src/legacy.]

# 🧪 Validation Specs
- [Automated testing, e.g., Running `npm run test:unit` must pass 100%.]
- [Edge behaviors, e.g., When inputs are empty, the API must return 400 Bad Request with a structured JSON error payload.]
```

4.3 Real Case Comparison: Traditional Prompt vs. Goal-Driven Specs

Suppose we need to write a **"Redis-based Rate-Limiting API Proxy Service"**.

❌ Traditional Process-Driven Prompt

> *"Please help me write an API proxy with Express. First, import express and express-rate-limit. Then configure rate-limiting, setting windowMs to 15 minutes and max to 100. Then write a route `/api/proxy` using axios to request the third-party API `https://api.github.com`. If successful, return the data; if it fails, return a 500 error. Make sure to include the Authorization Bearer Token in the headers."*

✅ Goal-Driven Specs (Recommended by Real-World Product Talk)

```
# 🎯 Goal
Implement an Express API proxy route that forwards all incoming requests to the GitHub API.

# 🛑 Constraints
- Must use Redis as the rate-limiting data source (no memory-based limiting) to support multi-instance deployments.
- Limit the proxy request timeout strictly to 3000ms to prevent hanging the main thread.
- Never write the GitHub token into the codebase or logs. It must be safely fetched from `process.env.GH_TOKEN`.
```

```
# 🧪 Validation Specs
- Return a 429 Too Many Requests status when requests exceed 60 requests per minute from a single IP.
- Return a 504 Gateway Timeout status with a structured JSON response on timeout or proxy network errors.
```

4.4 Codex Reasoning Process for Specs

Once you throw these specs to Codex, its internal Chain of Thought (CoT) will operate like this:

```
graph TD
    A["Parse Specs Goal"] --> B{"Analyze Constraints"}
    B -->|"Hard Constraint: Redis Limiting"| C["Decide to import ioredis and rate-limit-redis"]
    B -->|"Hard Constraint: 3s Timeout"| D["Configure timeout parameter in Axios/Fetch"]
    B -->|"Security Constraint: Secret Isolation"| E["Import dotenv and write TypeScript declarations"]
    C & D & E --> F["Plan code structure and write implementation"]
    F --> G{"Compare with Validation Specs"}
    G -->|"Simulate 429 scenario"| H["Write Redis mock test cases"]
    G -->|"Simulate 3s timeout"| I["Write latency API Mock and run unit tests"]
    H & I --> J["Output final code and test reports"]
```

You will notice that Codex automatically handles edge cases, such as Redis connection retries and error-catching on timeout—things that previously required hundreds of words of manual instruction.

Delegate logic planning to the AI, but keep verification standards firmly in your own hands. This is the most efficient human-machine collaboration method in the AI era.

Ch.05 Defining the CAP Protocol: Building Your Project's AGENTS.md Rule Compliance Layer

In the product development process, one of the most frustrating scenarios is fixing one bug only to introduce three new ones, or coding a new feature while completely disregarding the team's established coding standards.

In human-machine collaborative development, if Codex is left without boundary constraints, it too can become an overly eager and "destructive" employee. To place a tight rein on it, we need to establish an **`AGENTS.md`** file in the project root.

This is our **"agent collaboration constitution" (Codex Collaboration Protocol, CAP)**.

5.1 Why Do We Need `AGENTS.md` ?

While many developers in the Claude Code ecosystem use `CLAUDE.md` , in our Codex framework, we name it `AGENTS.md` . Its core value lies in:

1. **Immediate Context Restore**: Every time Codex starts, its first action is scanning the `AGENTS.md` file in the root directory. It instantly picks up the project's tech stack, directory structure, and collaboration constraints, eliminating the need for you to repeat instructions in chat.
2. **Anti-Corruption Layer**: It explicitly defines which directories and files are "read-only/off-limits," preventing Codex from unilaterally refactoring critical security modules (e.g., authentication, database schema).
3. **Command Execution Guardrails**: It specifies permissible test and deployment command options, avoiding the execution of destructive scripts by the AI.

5.2 The Four Core Sections of `AGENTS.md`

```markdown
# Project Fingerprint
- Tells the AI what kind of project this is and its core tech stack.

# Developer Commands
- Explicitly states the command lines for building, testing, and running database migrations.

# Styles & Architecture Patterns
- Dictates where files should be placed, size limits, and design patterns to use.

# Agent Boundary & Hard Rules
- Defines absolute forbidden paths. If touched, Codex must abort and ask for human verification.
```

5.3 Practice: A Production-Grade `AGENTS.md` Template

Here is a typical `AGENTS.md` specification for a full-stack SaaS project:

```markdown
# 🤖 Project: Aurora SaaS Core (AGENTS.md)

## 🧬 Project Fingerprint
- **Database**: PostgreSQL on Supabase.

## 💻 Developer Commands
- **Install**: `npm install` (Only run if package.json has changed)
- **Dev Server**: `npm run dev`
- **Lint Code**: `npm run lint`
- **Run Tests**: `npm run test`
- **DB Migration**: `npx prisma migrate dev` (Never run in production branch)

## 🎨 Styles & Architecture Patterns
- **Directory Structure**:
  - Components: Keep UI components inside `@/components/ui/` (Shadcn styled).
  - Business Logic: Custom React hooks must go to `@/hooks/`.
- **Formatting**:
  - Keep components modular. If a component exceeds 200 lines, decompose it.
  - All API routes must implement Zod schema validation for request body.
```

```
  - Return HTTP 400 for validation errors, 500 for internal uncaught errors.

## 🛑 Agent Boundary & Hard Rules
- **READ-ONLY Directories**:
  - Never modify files inside `src/app/api/auth/[...nextauth]` (OAuth Core).
  - Never alter `prisma/schema.prisma` without explicit human confirmation.
- **PR Rules**:
  - Before declaring a feature complete, run `npm run test` and `npm run lint`.
  - If tests fail, rollback the change immediately and report the error logs.
- **Security Check**:
  - Never commit raw `.env` files or API Keys. Use environment variables.
```

5.4 Founders' Advice: Making Constraints Actually Stick

In Codex, `AGENTS.md` is loaded by default and injected directly into the system context.

When you (or Codex itself) execute a task that deviates from the rules defined in `AGENTS.md`, Codex's telemetry mechanism will trigger a hard warning, pausing the current step and sending you an interruption confirmation:

```
⚠️ [Warning] Codex attempts to edit src/app/api/auth/[...nextauth]/route.ts.
This path is flagged as READ-ONLY in AGENTS.md.
Do you want to override this rule? (y/N)
```

With this layer of protection, you can confidently delegate large blocks of business logic implementation to Codex without having to constantly worry about whether it broke the underlying security layers.

Ch.06 Correcting Course: Supervising the CoT Reasoning Chain Like a Tech Lead

In traditional software development, when hiring a junior programmer, your biggest fear is not that they won't write code. Rather, it is that they will secretly work in isolation and ultimately hand you a pile of code that completely deviates from the business requirements, or worse, crashes the system.

This same challenge exists when using Codex driven by reasoning models (o-series / GPT-5.5). Although the AI possesses powerful reasoning capabilities, if its initial assumptions are incorrect, it will sprint down the wrong logical path and can even get stuck in an infinite "self-correction" loop.

This chapter teaches you how to look inside Codex's **Chain of Thought (CoT)** so you can intervene like a seasoned tech lead when the agent starts going off course.

6.1 Why Look Inside CoT (Chain of Thought)?

The biggest difference between reasoning models and standard LLMs (like GPT-4o) is that before outputting final code, the reasoning model "drafts" its reasoning and simulates the execution internally. This draft process is the Chain of Thought (CoT).

```
[User Request] ──> 1. Parse Goal & Constraints ──> 2. Plan Steps ──> 3. Run Tests ──>
4. Self-Correct ──> [Final Output]
                            └──────(Displayed in terminal as CoT Logs, your supervisor
viewport)──────┘
```

If you only inspect the final output, you are limited to a passive role. If you monitor the CoT logs in real-time, you can detect logical errors at step 2 or 3 and intervene early, avoiding wasted API tokens and time.

6.2 How to View and Interpret Codex's Real-Time CoT Logs

When running Codex, you can print its internal thinking process directly to the console by appending debugging flags to the CLI command:

```
# Start Codex and print detailed thinking process logs
codex --verbose --show-cot
```

Real-Time Logs Viewport Example

When Codex receives a task like "Fix connection timeout in the Redis rate limiter," the console will output a thought flow similar to this:

```
[Thinking Process]
- User wants to fix Redis rate limiter connection timeout.
- Checking existing implementation in src/lib/redis.ts...
- I see `redis = new Redis()` without retryStrategy.
- If Redis is down, it hangs the Express app. This violates the 3000ms timeout
constraint in AGENTS.md.
- Plan:
  1. Add `maxRetriesPerRequest: null` and `enableReadyCheck: false`.
  2. Implement custom retryStrategy up to 3 times.
  3. Write mock connection test to verify.
```

6.3 Identifying Typical AI "Infinite Loops"

From my hands-on experience running "Real-World Product Talk", I have identified three common infinite loops that AI gets trapped in. If you see these signs in the CoT, you must intervene immediately:

1. The Dependency Loop (Infinite `npm install`)

- **Symptom**: The AI attempts to use a new library, fails during installation, decides in the CoT to try another version, fails again, and then tries installing another similar package.
- **CoT Indicators**: `Error: Cannot resolve dependency ... Running npm install --legacy-peer-deps ...` repeating more than 3 times.

2. The Regression Loop (Self-Destructive Code Refactoring)

- **Symptom**: The AI edits file A, causing unit test B to fail; it modifies test B, which leads to module C throwing errors; it edits C, which breaks file A again.
- **CoT Indicators**: The AI constantly bounces back and forth between two or three files, and the test pass rate fluctuates repeatedly between 80% and 90%.

6.4 The Three-Step Intervention: Interruption, Correction, and Rollback

When you find the AI going off track or caught in a loop, do not just sit back. Intervene using these steps:

Step 1: Interrupt (`Ctrl + C` or `stop`)

Press `Ctrl + C` directly in the terminal or enter `stop` . This immediately freezes Codex's sandbox state, preventing it from consuming further tokens.

Step 2: Refine (`refine`)

After interrupting, Codex will enter an interactive command prompt mode. You can use the `refine` command to point out the logical blind spot directly:

```
# Point-to-point correction command
codex refine "You just attempted to install axios-retry. This project prohibits
installing any third-party HTTP libraries. Use native AbortController to implement
timeout retries instead."
```

Step 3: Rollback and Add Safeguards

If the AI has already altered the codebase beyond recognition, do not let it attempt to fix it. Roll back using git and append a hard rule to `AGENTS.md` :

```
# Revert the AI's erroneous modifications
git checkout -- src/lib/redis.ts
```

Then append this to [AGENTS.md]:

```
- Do not introduce external retry helper libraries for basic network timeout issues.
```

Ch.07 Closing the Visual Loop: Automated Auditing and Design Verification with Desktop Computer Use

In traditional UI fidelity reviews, the most time-consuming task for product managers and front-end developers is "pixel-eye" alignment verification: “This button seems shifted 4 pixels to the left.” “This modal breaks and covers key text on iPad dimensions.”

In the OpenAI Codex ecosystem, using **Computer Use** capabilities, the agent can not only write code but also "use eyes and hands" to directly operate your macOS/Windows desktop, launch a browser, interact with DevTools, and run visual audits.

This chapter teaches you how to configure and utilize Codex Desktop for automated UI visual verification.

7.1 Safety First: Sandbox Boundaries and Operation Bounding Boxes

Allowing an AI to operate your screen poses security risks. To prevent Codex from accidentally clicking your messaging apps or deleting system files due to misinterpretations, you must configure a **bounding box** restriction.

1. Defining Boundaries in Configuration

In the project root configuration, restrict Codex to access only a designated virtual display or window region:

```
{
  "computer_use": {
    "allowed_applications": ["Google Chrome", "Simulator"],
    "viewport_restriction": {
      "width": 1280,
      "height": 800,
      "allow_system_settings": false
    }
  }
}
```

2. Coordinate Targeting Mechanism

Codex operates your computer in a closed loop: "screenshot -> OCR/visual analysis -> return target x, y coordinates -> execute click/drag."

```
[Screenshot] ——> [Vision Model Analysis] ——> [Fetch Element Coordinates (x:450, y:230)] ——> [Click/Drag]
```

7.2 Visual-Driven UI Review in Practice: Figma Mockup Alignment

This is a classic and highly practical "Real-World Product Talk" workflow: **let Codex autonomously compare a Figma mockup screenshot with the rendered page in the browser, and automatically modify CSS to restore the design.**

🎯 Goal

Instruct Codex to compare the local webpage `/auth/login` with the designer's mockup `figma_login_mockup.png`, and automatically adjust the margins and font sizes of the login card on the page.

🛑 Constraints

- Only style adjustments through modifying the Tailwind classes in `src/app/login/page.tsx` are allowed.
- Modifying the DOM structure is prohibited.

🧪 Execution and Validation Specs

```
# 🎯 Goal
Compare and align browser rendering with figma_login_mockup.png.

# 🛑 Constraints
- Only use Tailwind utility classes in src/app/login/page.tsx.

# 🚀 Codex Execution Steps
1. Launch Google Chrome in headless or bounded window mode.
2. Navigate to http://localhost:3000/auth/login.
```

```
3. Take a screenshot of the login card region.
4. Perform pixel-diff and layout alignment checks against
/assets/figma_login_mockup.png.
5. Identify spacing discrepancy (Figma shows 32px padding-top, current implementation
has 16px).
6. Edit CSS, reload and verify.
```

7.3 Codex Automated Auditing Console Logs

When you run this task, Codex's Computer Use module generates logs similar to the following:

```
$ codex run-task compare-ui.task
[Task Started] Running visual review for /auth/login
[Step 1] Opening Google Chrome on http://localhost:3000/auth/login...
[Step 2] Taking screenshot. Saved to /tmp/screenshot_v1.png
[Step 3] Calling Vision Model (GPT-4o/o3) for image comparison.
    - Analysis: "Login card header text 'Welcome Back' font size is too small (approx
16px), should be 24px (text-2xl) based on figma mockup. Card padding-top is
insufficient."
[Step 4] Modifying src/app/login/page.tsx:
    - Target: Replace `className="text-base pt-4"` with `className="text-2xl pt-8"`
[Step 5] Refreshing Chrome...
[Step 6] Taking validation screenshot. Saved to /tmp/screenshot_v2.png
[Step 7] Vision model checks: "Layout matches mockup. Visual diff is within 1.5%
tolerance."
[Task Completed Successfully]
```

7.4 Best Practice: Responsive Multi-Device Inspection

Beyond single-resolution comparisons, you can command Codex to resize viewports for breakpoint audits:

```
# 📱 Mobile Viewport Inspection
1. Open DevTools in Chrome.
2. Simulate mobile viewport (iPhone 15 Pro: 393 x 852).
3. Verify that the login card does not overflow horizontally.
4. If the submit button falls below the screen fold, adjust padding-bottom to keep it
visible.
```

Through this closed-loop process, solo developers no longer need to resize browsers manually or refresh mobile devices repeatedly after modifying CSS. Codex can handle 90% of page alignment adjustments and cross-device compatibility testing, freeing up your visual energy.

Ch.08 Mobile Sentinel Workflows: 24/7 Remote Development and Orchestration

As an independent founder and product manager, your primary pursuit besides "high efficiency" is "freedom." You certainly do not want to be chained to your desk all day long, watching terminal consoles scroll compile logs.

Under the multi-surface collaborative OpenAI Codex ecosystem, we can construct a **"mobile watchdog workflow"**: While you are on the subway or at a café, Codex runs automated refactoring on your local or cloud server. The moment a build fails, or high-risk production deployment permissions are requested, your WeChat, Slack, or Telegram will instantly receive a notification card. You can reply with simple text commands from your phone to approve or guide the correction.

This chapter teaches you how to turn your mobile phone into the steering wheel for your remote Codex engineering army.

8.1 Architecture of the Mobile Watchdog Loop

We connect the local or cloud Codex agent to your phone through the following pipeline:

```
[Cloud Codex Agent] ──(Webhook)──> [Notification Gateway (Pusher/Slack API)] ──> [Mobile WeChat/Slack]
        ▲
│
        └───────────────(Reply "1" for Approve / "0" for Abort from phone)
─────────────────┘
```

8.2 Practice: Webhook Notification Setup on Build Failures

When Codex runs compile or test suites in a sandbox, we capture build logs via GitHub Actions and trigger alerts.

1. GitHub Actions Workflow Configuration (`.github/workflows/codex-watchdog.yml`)

Create the following workflow configuration in your project root. When a build fails, it posts a summary highlighting the critical failure nodes in the CoT reasoning chain directly to your phone:

```yaml
name: Codex Agent Watchdog

on:
  push:
    branches: [ main ]
  workflow_dispatch:

jobs:
  agent-build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Setup Node.js
        uses: actions/setup-node@v4
        with:
          node-version: 20

      - name: Run Codex Sandbox Compile & Test
```

```
        id: run_agent
        run: |
          # Simulate running Codex validation task
          npm ci
          npm run test || echo "STATUS=failed" >> $GITHUB_ENV

      - name: Push Fail Notice to Mobile
        if: env.STATUS == 'failed'
        run: |
          curl -X POST -H "Content-Type: application/json" \
            -d '{
              "event": "build_failed",
              "repo": "${{ github.repository }}",
              "commit": "${{ github.sha }}",
              "message": "Codex sandbox testing failed. Attention required on phone."
            }' \
            ${{ secrets.MOBILE_WEBHOOK_URL }}
```

8.3 Mobile Bidirectional Interaction and Approval

Receiving failure warnings is only the first step. The advanced usage is sending remote control commands to Codex directly from your phone.

1. Scenario: Production Deployment Approval Gate

When Codex passes all test suites and is ready to merge code into `main` and deploy to Vercel, it pauses and posts a card to your Slack or WeChat group:

```
🚨 [Codex Auth Requested]
Project: pmer-cn-saas
Action: Deploy to production (Vercel)
Change summary: Implemented Stripe subscription webhook in /api/stripe.
Tests: 12 passed, 0 failed.
[Directive Command]: Reply "1" to approve deployment, "0" to abort and roll back.
```

2. Node.js Gateway Script for Remote Execution

We deploy a minimalist gateway server script on the server behind `pmer.cn` to parse incoming messaging webhook payloads and communicate with the downstream agent instance:

```
// File: /Users/hunkwu/Desktop/ai/book/scratch/gateway.js
const express = require('express');
const { exec } = require('child_process');
const app = express();
app.use(express.json());

// Receive reply notifications from WeChat/Slack
app.post('/api/mobile-reply', (req, res) => {
  const { userMessage, user } = req.body;
```

```
  // Only allow owner hunkwu for remote control
  if (user !== 'hunkwu') {
    return res.status(403).json({ error: 'Unauthorized' });
  }

  if (userMessage === '1') {
    // Send signal to the background suspended Codex process, approving merge and
deploy
    exec('echo "approved" > /tmp/codex_deploy_signal', (err) => {
      if (err) return res.status(500).send('Error triggering deploy');
      res.json({ reply: '🚀 Deployment approved, production environment is going
live!' });
    });
  } else if (userMessage === '0') {
    // Force kill Codex process and roll back code
    exec('pkill -f codex && git checkout -- .', (err) => {
      res.json({ reply: '🛑 Deployment aborted, code safely rolled back to HEAD!' });
    });
  } else {
    res.json({ reply: '⚠️ Invalid instruction. Reply 1 (approve) or 0 (abort).' });
  }
});

app.listen(8080, () => console.log('Mobile gateway listening on port 8080'));
```

8.4 Founder's Mantra: Reclaiming Your Freedom

Many tech practitioners using AI tools end up behaving like "manual testing monkeys" and "human git commit triggers." AI edits code, the human refreshes the tab; AI returns an error, the human copies the trace and pastes it back to the chat.

The essence of the mobile watchtower workflow is detaching humans from constant, immediate waiting.

By delegating validation assertions to GitHub Actions, forwarding exceptions via mobile webhooks, and holding the deployment approval key on your mobile device, you can achieve the dream: **"enjoying your coffee while the product automatically evolves."**

Ch.09 Codebase Revitalization: Reverse Engineering and Progressive Decoupling of Legacy Systems

When working on solo projects or inheriting outsourced contracts, the biggest headache is not building from scratch. It is inheriting a legacy "spaghetti code" system left behind by previous developers, complete with zero documentation and zero unit tests. Every minor code modification feels like dancing in a minefield.

In "Real-World Product Talk", I often advise: **Do not blindly tear down and rewrite. By leveraging Codex's panoramic analysis and reasoning power, we can implement progressive decoupling and refactoring of legacy systems.**

This chapter teaches you how to direct Codex to act as your legacy codebase operator.

9.1 Step 1: Panoramic Reverse Engineering - Generating Codebase Topology

The first task when inheriting a messy project is **drawing the map**. Do not comb through the code yourself; let Codex reverse-engineer the project configuration for you.

1. Reverse-Engineering Database Schemas

If the project uses Prisma or raw SQL, you can ask Codex to reverse-engineer a Mermaid Entity-Relationship Diagram (ERD) directly from the schema file.

Dispatch the following task specs locally to Codex:

```
# 🎯 Goal
Analyze the database configuration of the current project and generate a Mermaid ERD
illustrating core table structures and foreign key relationships.

# 🛑 Constraints
- Only analyze files under /prisma/schema.prisma or /src/db/models/.
- Exclude temporary tables and third-party metadata tables.

# 🧪 Validation Specs
- Output a properly rendered Markdown Mermaid code block in
/docs/database_topology.md.
```

Codex will automatically parse table relations and generate an intuitive topology diagram, which is hundreds of times faster than manually auditing database relationships.

9.2 Step 2: Safety Guardrails - Writing Characterization Tests

The golden rule of refactoring legacy code is: **Verify that old behaviors are not broken before writing new logic.** We must instruct Codex to automatically write **characterization tests** for existing key endpoints (e.g., checkout payments, OAuth callbacks) to serve as safety boundaries.

Practice: Dispatching Characterization Test Instructions to Codex

```
# 🎯 Goal
Write unit tests for the src/pages/api/checkout.ts endpoint to capture current request
behaviors and response payloads.

# 🛑 Constraints
- Do not modify any logic in src/pages/api/checkout.ts.
- Run tests using Jest / Vitest inside the local sandbox.

# 🧪 Validation Specs
- Cover at least three scenarios: successful product checkout, out-of-stock errors,
and unauthenticated access block.
- Ensure the test suite pass rate is 100%.
```

With these characterization tests serving as the AI's validation guardrails, any subsequent refactoring changes that break legacy logic will be instantly caught during compilation and testing inside the sandbox.

9.3 Step 3: Minimally Invasive Surgery - Implementing Progressive Decoupling

Equipped with our "map" (topology diagram) and "shield" (characterization tests), we can now direct Codex to perform the actual refactoring.

1. Decoupling Fat Controllers

Suppose we have a 500-line API route that mixes authentication, discount calculations, emails, inventory updates, and logging. We can command Codex to extract this logic:

```
# 🎯 Goal
Extract the "discount calculation logic" from src/pages/api/checkout.ts into a
standalone service class src/services/discountService.ts.

# 🛑 Constraints
- Ensure the endpoint input and output schemas remain completely unchanged.
- Strictly avoid affecting other core logic within checkout.ts.

# 🧪 Validation Specs
- Run `npm run test` and ensure the previously written checkout characterization tests
pass 100%.
- Run `npm run lint` and verify there are no syntax or formatting errors.
```

2. Local Validation and Automated Rollbacks

As Codex begins the refactoring task, it will modify the code and run the test suite inside the sandbox. If the tests fail, it will analyze its CoT thinking chain to determine whether the refactoring code was flawed or if the characterization tests were outdated.

Using this minimally invasive refactoring cycle (Extract Logic -> Run Tests -> Rollback on Error -> Readjust), you can refactor a chaotic, unmaintainable legacy codebase of tens of thousands of lines into a clean, modular, and tested system in just a few hours.

Do not fear legacy spaghetti code. Use goal-driven specs and characterization tests as your shield, and let Codex handle the heavy lifting of refactoring.

Ch.10 Monetization in Practice: Shipping a Commercial SaaS MVP in 2 Hours

As an independent developer (Indie Hacker) or micro-startup, your key milestone is not building a "perfect architecture." It is **"receiving your first payment."** Many developers spend far too much time repeatedly configuring boilerplate project templates and delaying their actual launch.

10.1 Initialization and Database Schema Design

Our goal is to build a subscription-based AI translation service. First, create AGENTS.md in the project root to enforce strict boundaries. Then, instruct Codex to generate the core database models.

1. Designing the Prisma Schema (Database Entity Modeling)

Dispatch the following goal-driven specs to Codex:

```
# 🎯 Goal
Write Prisma database models supporting User, Subscription, and TranslationRecord
entities.

# 🛑 Constraints
- Use PostgreSQL (connected via Supabase) as the database provider.
- Subscription status must be an Enum type containing ACTIVE, CANCELED, and EXPIRED.

# 🧪 Validation Specs
- Running `npx prisma validate` must return no syntax or definition errors.
```

Codex will automatically output a standard `schema.prisma` file containing foreign key relationships, cascade deletes, and database indexes:

```
// File: prisma/schema.prisma
datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

generator client {
  provider = "prisma-client-js"
}

enum SubscriptionStatus {
  ACTIVE
  CANCELED
  EXPIRED
}

model User {
  id             String               @id @default(uuid())
  email          String               @unique
  createdAt      DateTime             @default(now())
  subscription   Subscription?
  records        TranslationRecord[]
}

model Subscription {
  id             String               @id @default(uuid())
  userId         String               @unique
  stripeSubId    String               @unique
  status         SubscriptionStatus
  priceId        String
  currentPeriodEnd DateTime
  user           User                 @relation(fields: [userId], references: [id],
onDelete: Cascade)
}
```

```
model TranslationRecord {
  id             String              @id @default(uuid())
  userId         String
  sourceText     String
  translatedText String
  createdAt      DateTime            @default(now())
  user           User                @relation(fields: [userId], references: [id],
onDelete: Cascade)
}
```

10.2 Integrating Stripe Subscriptions and Webhook Handlers

Practice: Dispatching Webhook Route Specifications to Codex

```markdown
# 🎯 Goal

# 🛑 Constraints
- Verify the signature of incoming webhook requests using
`stripe.webhooks.constructEvent` to prevent forgery attacks.
- Update user subscription status in the database on receiving
`checkout.session.completed` or `invoice.payment_succeeded` events.
- Do not parse the request body as JSON. Stripe verification requires the raw binary
buffer.

# 🧪 Validation Specs
- Write automated mock requests verifying that invalid signatures return HTTP 400 Bad
Request, while valid sessions return HTTP 200 OK.
```

```typescript
// File: src/app/api/webhooks/stripe/route.ts
import Stripe from 'stripe';
import { prisma } from '@/lib/prisma';

const stripe = new Stripe(process.env.STRIPE_SECRET_KEY!, {
  apiVersion: '2024-12-18.acacia',
});

export async function POST(req: Request) {
  const body = await req.text();
  const signature = req.headers.get('stripe-signature')!;

  let event: Stripe.Event;

  try {
    event = stripe.webhooks.constructEvent(
      body,
      signature,
      process.env.STRIPE_WEBHOOK_SECRET!
```

```
    );
  } catch (err: any) {
  }

  // Handle state transition
  if (event.type === 'checkout.session.completed') {
    const session = event.data.object as Stripe.Checkout.Session;
    const stripeSubId = session.subscription as string;
    const customerEmail = session.customer_details?.email!;

    // Asynchronously update database state
    await prisma.subscription.upsert({
      where: { stripeSubId },
      update: { status: 'ACTIVE' },
      create: {
        stripeSubId,
        status: 'ACTIVE',
        priceId: session.metadata?.priceId || 'default',
        currentPeriodEnd: new Date(Date.now() + 30 * 24 * 60 * 60 * 1000), //
Temporarily set to 30 days for demonstration
        user: { connect: { email: customerEmail } }
      }
    });
  }

}
```

10.3 Local Debugging: Using the Sandbox Mock for E2E Validation

Debugging Stripe locally requires the Stripe CLI to forward webhooks. Using the network penetration techniques covered in Ch.03, we can configure our local environment:

1. Run the Stripe webhook forwarding command locally:

   ```
   stripe listen --forward-to localhost:3000/api/webhooks/stripe
   ```

2. Add the `whsec_xxx` webhook signing secret printed by the console to your local `.env` file and instruct Codex to run the verification tests.

By following this loop (Spec Definition -> AI Coding -> Sandbox Validation -> Real-World Stripe Integration Test), independent developers can compress what normally takes two days of integration struggle down to under 30 minutes.

The value of a commercial MVP lies in speed. Enforce strict boundaries on the AI to swap for maximum speed-to-market.

Ch.11 Mobile Extension: Expo Cross-Platform App Development and Cloud Packaging

After completing a web-based SaaS, many independent developers hope to extend their reach to mobile. However, in traditional mobile development (React Native or Flutter), the most time-consuming part is often the complex local environment setup: iOS certificate management, Android Gradle errors, Cocoapods version conflicts. This local environment hell often deters developers.

In "Real-World Product Talk", I firmly believe: **"Cloud compilation and packaging (EAS) is the only viable path for independent developers to build native mobile apps."** Combined with Codex's automated compilation error diagnostics, you can entirely skip the tedious configurations of local Xcode/Android Studio and directly package a native App ready for submission.

This chapter teaches you how to direct Codex to handle all of this autonomously.

11.1 Rapid Expo Project Initialization

To use EAS (Expo Application Services) for configuration-free cloud packaging, we first need to establish a connection between the sandbox and the local environment.

1. Writing App Initialization Specs

Issue the following specs command to Codex to generate a standard Expo TypeScript project:

```
# 🎯 Goal
Initialize a React Native Expo project using TypeScript.

# 🛑 Constraints
- Use Expo Router v3 to implement file-system-based routing.
- Integrate the Tailwind CSS styled library NativeWind.

# 🧪 Validation Specs
- Running `npx expo lint` must return zero errors.
```

Codex will automatically pull the latest Expo template and generate the basic directory structure:

```
+-- src/
|   +-- app/
|   |   +-- index.tsx          # App Home Page
|   |   +-- _layout.tsx        # Global Routing Navigation Layout
|   +-- components/
|   +-- hooks/
+-- app.json                   # Expo Core Configuration File
+-- package.json
```

11.2 Resolving Mobile Obstacles: Native Module Conflicts

The most feared scenario in React Native development is upgrading or introducing a third-party library that contains native modules (such as Camera or File System access). This frequently leads to build failures in iOS Podfile or Android build.gradle.

Under Vibe Coding, when Codex installs dependencies and encounters native errors, we should guide it to follow the troubleshooting strategy below.

Practice: Guiding Codex to Troubleshoot Podfile Failures

If Codex fails to compile iOS native modules inside the sandbox, its CoT logs will display a warning similar to: `[Error] [Cocoapods] Auto-linking failed for react-native-reanimated.`

At this point, use the `refine` command in the CLI to add specific constraints:

```
codex refine "Reinstall the package using 'npx expo install' to match SDK versions. Never use standard 'npm install' to install third-party packages containing native mobile code."
```

> 💡 ***Founder's Advice***: *The greatest strength of Expo SDK is its built-in dependency alignment mechanism (`npx expo install`). Whenever a native package throws an error, force Codex to use Expo's native package command to install dependencies, which automatically resolves 90% of version conflicts.*

11.3 EAS Build Cloud Packaging and Certificate Automation

In traditional App packaging pipelines, applying for Apple developer certificates and generating Provisioning Profiles can stall beginners for days. Now, using EAS, we only need to provide developer credentials to Codex, and it handles the entire setup on the cloud automatically.

1. Configuring `eas.json` (EAS Cloud Build Config)

Have Codex generate the cloud build environment configurations for production and testing.

`eas.json` configuration file:

```json
{
  "cli": {
    "version": ">= 9.0.0"
  },
  "build": {
    "development": {
      "developmentClient": true,
      "distribution": "internal"
    },
    "preview": {
      "distribution": "internal"
    },
    "production": {
      "ios": {
        "simulator": false
      }
    }
  }
}
```

2. Instructing Codex to Monitor Cloud Build Logs

Trigger the EAS build task from the local terminal:

```
# Start EAS iOS build task and pipe logs to Codex
eas build --platform ios --profile production --non-interactive | codex watch-eas
```

During the cloud compilation process, if a Provisioning Profile mismatch or certificate expiration occurs, Codex will analyze the cloud build logs in its CoT thinking chain in real-time, reconnect to sync the developer console certificate state, and auto-correct.

Ultimately, it will output a QR code. You only need to scan it with your phone to download and install the preview App.

Native mobile development no longer requires a bulky IDE setup. Move packaging to the cloud with Expo + EAS, and let Codex act as your safety net.

Ch.12 The Final Frontier: Building an Automated Growth Flywheel for a One-Person SaaS

On my WeChat public account "Real-World Product Talk" and pmer.cn, I have written numerous articles about "independent development and side hustles." I have observed that the most common trap developers fall into is: **spending 99% of their energy optimizing code, but only 1% of their energy finding users.**

No matter how elegant your code is or how perfect your architecture config is, as long as nobody uses it, it is just a pretty ornament. In the AI era, we must not only let Codex help us "manufacture products," but also let it help us "spin the commercial flywheel."

As the conclusion of this book, let us discuss how to leverage AI to achieve automated marketing and organic growth.

12.1 Automated Traffic Pipeline for a "One-Person SaaS"

For a healthy indie project, traffic acquisition (SEO, social media, cold emailing) should function as an automated conveyor belt just like its codebase.

Practice: Directing Codex to Autonomously Maintain SEO Blogs and Sitemaps

We can write a Node.js script that prompts Codex daily to automatically scrape industry posts based on trending keywords, summarize them into high-quality blogs, and update static routes and the XML sitemap, thereby capturing long-tail search traffic.

Establish the site mapping script specs in your project:

```markdown
# 🎯 Goal

# 🛑 Constraints
- Must include the paths of all newly generated Markdown articles under the `/blog` directory.
- Ensure change frequency (`changefreq`) and priority parameters conform to Google schema recommendations.

# 🧪 Validation Specs
```

```
- Running `node scripts/generate-sitemap.js` must generate an XML file that parses
normally in browsers with zero missing formatting tags.
```

Codex will automatically output a standard sitemap generation script:

```javascript
// File: scripts/generate-sitemap.js
const fs = require('fs');
const path = require('path');

const BASE_URL = 'https://pmer.cn';
const pagesDir = path.join(__dirname, '../src/app');
const blogDir = path.join(__dirname, '../content/blog');

function getBlogSlugs() {
  if (!fs.existsSync(blogDir)) return [];
  return fs.readdirSync(blogDir)
    .filter(file => file.endsWith('.md'))
    .map(file => `/blog/${file.replace('.md', '')}`);
}

function generate() {
  const staticPages = ['/', '/auth/login', '/features'];
  const blogPages = getBlogSlugs();
  const allUrls = [...staticPages, ...blogPages];

  const xml = `<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  ${allUrls.map(url => `
  <url>
    <loc>${BASE_URL}${url}</loc>
    <changefreq>daily</changefreq>
    <priority>${url === '/' ? '1.0' : '0.8'}</priority>
  </url>`).join('')}
</urlset>`;

  fs.writeFileSync(path.join(__dirname, '../public/sitemap.xml'), xml);
  console.log('✅ Sitemap.xml generated successfully!');
}

generate();
```

12.2 Hooking Up Data Sentinels for Daily Business Reports

To keep yourself sensitive to cash flow dynamics, you can ask Codex to write a lightweight telemetry script that fetches Stripe's yesterday earnings and new user registrations from Supabase every morning and broadcasts a report card straight to your phone.

Daily Data Report Script (Node.js)

```javascript
// File: scripts/daily-report.js
const { PrismaClient } = require('@prisma/client');
const prisma = new PrismaClient();
const axios = require('axios');

async function sendReport() {
  // Get the count of new users registered yesterday
  const yesterday = new Date(Date.now() - 24 * 60 * 60 * 1000);
  const userCount = await prisma.user.count({
    where: { createdAt: { gte: yesterday } }
  });

  // Get the count of active subscriptions
  const activeSubs = await prisma.subscription.count({
    where: { status: 'ACTIVE' }
  });

  const reportText = `📊 [Real-World Product Briefing]\nNew registered users yesterday: ${userCount}\nTotal active subscriptions: ${activeSubs}\n—— Keep going!`;

  // Send notification to WeChat/Slack webhook
  await axios.post(process.env.MOBILE_WEBHOOK_URL, {
    msgtype: 'text',
    text: { content: reportText }
  });
}

sendReport();
```

12.3 Final Lesson: The Only Barrier in the AI-Native Era

When anyone can generate thousands of lines of code in two hours, build a native mobile app, and hook up automated marketing pipelines, **technology itself becomes completely commoditized**.

In this new era of "infinite code," the ultimate moat for independent developers and product managers is no longer knowing how to use a specific framework, but rather:

- **Your deep empathy for the user's actual pain points (User Empathy)**.
- **Your domain expertise accumulated over years in a specific industry (Domain Knowledge)**.
- **Your execution ability to turn AI-Native tools (like Codex) into your own leverage, shipping products and completing the commercial loop quickly.**

Do not be a mere code typist. Be the one who defines the problems, holds the reins, and resolves real-world pain points.

Thank you for reading the *Codex Blue Book*. Now, configure your `AGENTS.md` in your project root, run your first `codex` command, and go write your own business story!
